

ApexPlanet Data Analytics Internship

Task 5 – Final Reporting, Automation & Presentation

Project: Supermarket Sales Data Analytics

Prepared using the actual project outputs and analysis.

1. Executive Summary

This project demonstrates an end-to-end supermarket sales analytics workflow using Python, SQL, Power BI and Tableau. The workflow covers data preparation, exploratory data analysis, database analysis, dashboard development and an automated Python reporting pipeline. The final automation loads the raw CSV, checks data quality, removes duplicate records, converts the date field, creates a processed dataset and exports summary reports.

2. Problem Statement and Objectives

The objective is to transform supermarket transaction data into useful business information that can support sales monitoring and comparison across branches and cities.

- Clean and validate the supermarket sales dataset.
- Explore sales, quantity, rating and customer-related information.
- Perform SQL-based analysis using the supermarket database.
- Create interactive Power BI and Tableau dashboards.
- Automate repeatable cleaning and reporting tasks with Python.
- Produce a final set of documented analytical outputs.

3. Data Source and Methodology

The project uses the supermarket sales CSV dataset already present in the project repository. The dataset contains 10 transaction records and 13 columns, including Invoice_ID, Date, Branch, City, Customer_Type, Gender, Product_Line, Unit_Price, Quantity, Tax_5%, Total, Payment and Rating.

- Raw CSV loading using Pandas.
- Missing-value and duplicate checks.
- Duplicate removal during the automated cleaning pipeline.
- Date conversion for analysis.
- Exploratory statistical summary.
- Branch and city sales aggregation.
- Dashboard visualization using Power BI and Tableau.
- Automated CSV report generation.

4. Data Quality Results

Metric	Result
Records	10
Columns	13
Missing values	0
Duplicate rows	0
Average quantity	3.0
Average rating	8.7

5. Key Findings

- Total sales across the 10 records are 10,500.
- Average sales per transaction are 1,050.
- Sales range from 525 to 1,680 per transaction.
- Average customer rating is 8.7, indicating strong ratings within this dataset.
- Branch A has the highest sales at 3,832.5.
- Branch C follows with sales of 3,727.5.
- Branch B has the lowest sales at 2,940.0.
- The city ranking matches the branch ranking: Coimbatore, Bangalore, then Chennai.

6. Statistical Analysis

Measure	Value
Total Sales	10,500.0
Average Sales	1,050.0
Average Quantity	3.0
Average Rating	8.7
Minimum Sales	525.0
Maximum Sales	1,680.0

7. Branch and City Sales

Branch	Sales
A	3,832.5
C	3,727.5
B	2,940.0

City	Sales
Coimbatore	3,832.5
Bangalore	3,727.5
Chennai	2,940.0

8. Dashboard Deliverables

The project includes a Power BI dashboard and a Tableau dashboard. The dashboards contain KPI-style views for total sales, total orders and average rating, together with sales-by-city analysis.

Power BI file: Task3_PowerBI_Dashboard.pbix

Tableau file: Task4_Tableau_Dashboard.twbx

9. Automation Pipeline

The Task 5 automation script is stored as **scripts/automation_pipeline.py**. It automatically loads the raw dataset, checks data quality, removes duplicates, converts the Date column, saves the processed dataset, calculates summary statistics and generates sales-by-branch and sales-by-city CSV reports.

Generated outputs:

- data/processed/cleaned_supermarket_sales.csv
- reports/summary_statistics.csv
- reports/sales_by_branch.csv
- reports/sales_by_city.csv

10. Recommendations

- Monitor Branch B/Chennai sales closely because it has the lowest sales in the current sample.
- Use branch and city comparisons in periodic management reviews.
- Continue tracking customer ratings alongside sales to identify changes in customer experience.
- Run the automated pipeline whenever new transaction data becomes available.
- Use the Power BI and Tableau dashboards for management-friendly visual monitoring.

11. Limitations

- The current dataset contains only 10 transaction records, so findings should not be generalized to a larger supermarket population.
- The available dataset does not provide enough observations for robust predictive modelling.
- Dashboard and statistical conclusions are descriptive and based only on the supplied records.

12. Future Scope

- Add a larger and continuously updated transaction dataset.
- Schedule the automation pipeline using Windows Task Scheduler or another scheduler.
- Add richer time-based sales trends and product-level analysis.

- Extend the dashboard with additional filters and KPIs.
- Evaluate predictive models when sufficient historical data becomes available.

13. Conclusion

The project demonstrates a complete analytics workflow from raw supermarket transaction data to cleaned data, exploratory analysis, SQL analysis, interactive dashboards and automated reporting. The final automation pipeline improves repeatability and produces consistent analytical outputs for the project.